Corollary 3.37. *A closed manifold of odd dimension has Euler characteristic zero.*

Proof: Let M be a closed n-manifold. If M is orientable, we have $\operatorname{rank} H_i(M;\mathbb{Z}) = \operatorname{rank} H^{n-i}(M;\mathbb{Z})$, which equals $\operatorname{rank} H_{n-i}(M;\mathbb{Z})$ by the universal coefficient theorem. Thus if n is odd, all the terms of $\sum_i(-1)^i \operatorname{rank} H_i(M;\mathbb{Z})$ cancel in pairs.

If M is not orientable we apply the same argument using $\mathbb{Z}_2$ coefficients, with $\operatorname{rank} H_i(M;\mathbb{Z})$ replaced by $\dim H_i(M;\mathbb{Z}_2)$, the dimension as a vector space over $\mathbb{Z}_2$, to conclude that $\sum_i(-1)^i \dim H_i(M;\mathbb{Z}_2) = 0$. It remains to check that this alternating sum equals the Euler characteristic $\sum_i(-1)^i \operatorname{rank} H_i(M;\mathbb{Z})$. We can do this by using the isomorphisms $H_i(M;\mathbb{Z}_2) \approx H^i(M;\mathbb{Z}_2)$ and applying the universal coefficient theorem for cohomology. Each $\mathbb{Z}$ summand of $H_i(M;\mathbb{Z})$ gives a $\mathbb{Z}_2$ summand of $H^i(M;\mathbb{Z}_2)$. Each $\mathbb{Z}_m$ summand of $H_i(M;\mathbb{Z})$ with m even gives $\mathbb{Z}_2$ summands of $H^i(M;\mathbb{Z}_2)$ and $H^{i+1}(M,\mathbb{Z}_2)$, whose contributions to $\sum_i(-1)^i \dim H_i(M;\mathbb{Z}_2)$ cancel. And $\mathbb{Z}_m$ summands of $H_i(M;\mathbb{Z})$ with m odd contribute nothing to $H^*(M;\mathbb{Z}_2)$. $\square$

Connection with Cup Product

Cup and cap product are related by the formula

$$(*) \qquad \psi(\alpha \frown \varphi) = (\varphi \smile \psi)(\alpha)$$

for $\alpha \in C_{k+\ell}(X;R)$, $\varphi \in C^k(X;R)$, and $\psi \in C^\ell(X;R)$. This holds since for a singular $(k+\ell)$-simplex $\sigma:\Delta^{k+\ell}\to X$ we have

$$\begin{aligned}\psi(\sigma \frown \varphi) &= \psi(\varphi(\sigma|[v_0,\cdots,v_k])\sigma|[v_k,\cdots,v_{k+\ell}]) \\ &= \varphi(\sigma|[v_0,\cdots,v_k])\psi(\sigma|[v_k,\cdots,v_{k+\ell}]) = (\varphi \smile \psi)(\sigma)\end{aligned}$$

The formula $(*)$ says that the map $\varphi\smile : C^\ell(X;R)\to C^{k+\ell}(X;R)$ is equal to the map $\operatorname{Hom}_R(C_\ell(X;R),R)\to\operatorname{Hom}_R(C_{k+\ell}(X;R),R)$ dual to $\frown\varphi$. Passing to homology and cohomology, we obtain the commutative diagram at the right. When the maps h are isomorphisms, for example when R is a field or when $R=\mathbb{Z}$ and the homology groups of X are free, then the map $\varphi\smile$ is the dual of $\frown\varphi$. Thus in these cases cup and cap product determine each other, at least if one assumes finite generation so that cohomology determines homology as well as vice versa. However, there are examples where cap and cup products are not equivalent when $R=\mathbb{Z}$ and there is torsion in homology.

$$\begin{array}{ccc} H^\ell(X;R) & \xrightarrow{h} & \operatorname{Hom}_R(H_\ell(X;R),R) \\ \big\downarrow{\scriptstyle \varphi\smile} & & \big\downarrow{\scriptstyle (\frown\varphi)^*} \\ H^{k+\ell}(X;R) & \xrightarrow{h} & \operatorname{Hom}_R(H_{k+\ell}(X;R),R) \end{array}$$

By means of the formula $(*)$, Poincaré duality has nontrivial implications for the cup product structure of manifolds. For a closed R-orientable n-manifold M, consider the cup product pairing

$$H^k(M;R)\times H^{n-k}(M;R)\longrightarrow R, \qquad (\varphi,\psi)\mapsto(\varphi\smile\psi)[M]$$

Such a bilinear pairing $A \times B \to R$ is said to be **nonsingular** if the maps $A \to \mathrm{Hom}(B,R)$ and $B \to \mathrm{Hom}(A,R)$, obtained by viewing the pairing as a function of each variable separately, are both isomorphisms.

Proposition 3.38. *The cup product pairing is nonsingular for closed R-orientable manifolds when R is a field, or when $R = \mathbb{Z}$ and torsion in $H^*(M;\mathbb{Z})$ is factored out.*

Proof: Consider the composition

$$H^{n-k}(M;R) \xrightarrow{h} \mathrm{Hom}_R(H_{n-k}(M;R),R) \xrightarrow{D^*} \mathrm{Hom}_R(H^k(M;R),R)$$

where h is the map appearing in the universal coefficient theorem, induced by evaluation of cochains on chains, and D^* is the Hom-dual of the Poincaré duality map $D : H^k \to H_{n-k}$. The composition D^*h sends $\psi \in H^{n-k}(M;R)$ to the homomorphism $\varphi \mapsto \psi([M] \frown \varphi) = (\varphi \smile \psi)[M]$. For field coefficients or for integer coefficients with torsion factored out, h is an isomorphism. Nonsingularity of the pairing in one of its variables is then equivalent to D being an isomorphism. Nonsingularity in the other variable follows by commutativity of cup product. $\square$

Corollary 3.39. *If M is a closed connected orientable n-manifold, then for each element $\alpha \in H^k(M;\mathbb{Z})$ of infinite order that is not a proper multiple of another element, there exists an element $\beta \in H^{n-k}(M;\mathbb{Z})$ such that $\alpha \smile \beta$ is a generator of $H^n(M;\mathbb{Z}) \approx \mathbb{Z}$. With coefficients in a field the same conclusion holds for any $\alpha \neq 0$.*

Proof: The hypotheses on α mean that it generates a $\mathbb{Z}$ summand of $H^k(M;\mathbb{Z})$. There is then a homomorphism $\varphi : H^k(M;\mathbb{Z}) \to \mathbb{Z}$ with $\varphi(\alpha) = 1$. By the nonsingularity of the cup product pairing, φ is realized by taking cup product with an element $\beta \in H^{n-k}(M;\mathbb{Z})$ and evaluating on $[M]$, so $\alpha \smile \beta$ generates $H^n(M;\mathbb{Z})$. The case of field coefficients is similar. $\square$

Example 3.40: Projective Spaces. The cup product structure of $H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ as a truncated polynomial ring $\mathbb{Z}[\alpha]/(\alpha^{n+1})$ with $|\alpha| = 2$ can easily be deduced from this as follows. The inclusion $\mathbb{C}\mathrm{P}^{n-1} \hookrightarrow \mathbb{C}\mathrm{P}^n$ induces an isomorphism on H^i for $i \le 2n-2$, so by induction on n, $H^{2i}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ is generated by α^i for $i < n$. By the corollary, there is an integer m such that the product $\alpha \smile m\alpha^{n-1} = m\alpha^n$ generates $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$. This can only happen if $m = \pm 1$, and therefore $H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \approx \mathbb{Z}[\alpha]/(\alpha^{n+1})$. The same argument shows $H^*(\mathbb{H}\mathrm{P}^n;\mathbb{Z}) \approx \mathbb{Z}[\alpha]/(\alpha^{n+1})$ with $|\alpha| = 4$. For $\mathbb{R}\mathrm{P}^n$ one can use the same argument with $\mathbb{Z}_2$ coefficients to deduce that $H^*(\mathbb{R}\mathrm{P}^n;\mathbb{Z}_2) \approx \mathbb{Z}_2[\alpha]/(\alpha^{n+1})$ with $|\alpha| = 1$. The cup product structure in infinite-dimensional projective spaces follows from the finite-dimensional case, as we saw in the proof of Theorem 3.12.

Could there be a closed manifold whose cohomology is additively isomorphic to that of $\mathbb{C}\mathrm{P}^n$ but with a different cup product structure? For $n = 2$ the answer is no since duality implies that the square of a generator of H^2 must be a generator of

H^4. For $n = 3$, duality says that the product of generators of H^2 and H^4 must be a generator of H^6, but nothing is said about the square of a generator of H^2. Indeed, for $S^2 \times S^4$, whose cohomology has the same additive structure as $\mathbb{C}P^3$, the square of the generator of $H^2(S^2 \times S^4;\mathbb{Z})$ is zero since it is the pullback of a generator of $H^2(S^2;\mathbb{Z})$ under the projection $S^2 \times S^4 \to S^2$, and in $H^*(S^2;\mathbb{Z})$ the square of the generator of H^2 is zero. More generally, an exercise for §4.D describes closed 6-manifolds having the same cohomology groups as $\mathbb{C}P^3$ but where the square of the generator of H^2 is an arbitrary multiple of a generator of H^4.

Example 3.41: Lens Spaces. Cup products in lens spaces can be computed in the same way as in projective spaces. For a lens space L^{2n+1} of dimension $2n+1$ with fundamental group $\mathbb{Z}_m$, we computed $H_i(L^{2n+1};\mathbb{Z})$ in Example 2.43 to be $\mathbb{Z}$ for $i = 0$ and $2n+1$, $\mathbb{Z}_m$ for odd $i < 2n+1$, and 0 otherwise. In particular, this implies that L^{2n+1} is orientable, which can also be deduced from the fact that L^{2n+1} is the orbit space of an action of $\mathbb{Z}_m$ on S^{2n+1} by orientation-preserving homeomorphisms, using an exercise at the end of this section. By the universal coefficient theorem, $H^i(L^{2n+1};\mathbb{Z}_m)$ is $\mathbb{Z}_m$ for each $i \le 2n+1$. Let $\alpha \in H^1(L^{2n+1};\mathbb{Z}_m)$ and $\beta \in H^2(L^{2n+1};\mathbb{Z}_m)$ be generators. The statement we wish to prove is:

$$H^j(L^{2n+1};\mathbb{Z}_m) \text{ is generated by } \begin{cases} \beta^i & \text{for } j = 2i \\ \alpha\beta^i & \text{for } j = 2i+1 \end{cases}$$

By induction on n we may assume this holds for $j \le 2n-1$ since we have a lens space $L^{2n-1} \subset L^{2n+1}$ with this inclusion inducing an isomorphism on H^j for $j \le 2n-1$, as one sees by comparing the cellular chain complexes for L^{2n-1} and L^{2n+1}. The preceding corollary does not apply directly for $\mathbb{Z}_m$ coefficients with arbitrary m, but its proof does since the maps $h: H^i(L^{2n+1};\mathbb{Z}_m) \to \mathrm{Hom}(H_i(L^{2n+1};\mathbb{Z}_m),\mathbb{Z}_m)$ are isomorphisms. We conclude that $\beta \smile k\alpha\beta^{n-1}$ generates $H^{2n+1}(L^{2n+1};\mathbb{Z}_m)$ for some integer k. We must have k relatively prime to m, otherwise the product $\beta \smile k\alpha\beta^{n-1} = k\alpha\beta^n$ would have order less than m and so could not generate $H^{2n+1}(L^{2n+1};\mathbb{Z}_m)$. Then since k is relatively prime to m, $\alpha\beta^n$ is also a generator of $H^{2n+1}(L^{2n+1};\mathbb{Z}_m)$. From this it follows that β^n must generate $H^{2n}(L^{2n+1};\mathbb{Z}_m)$, otherwise it would have order less than m and so therefore would $\alpha\beta^n$.

The rest of the cup product structure on $H^*(L^{2n+1};\mathbb{Z}_m)$ is determined once α^2 is expressed as a multiple of β. When m is odd, the commutativity formula for cup product implies $\alpha^2 = 0$. When m is even, commutativity implies only that α^2 is either zero or the unique element of $H^2(L^{2n+1};\mathbb{Z}_m) \approx \mathbb{Z}_m$ of order two. In fact it is the latter possibility which holds, since the 2-skeleton L^2 is the circle L^1 with a 2-cell attached by a map of degree m, and we computed the cup product structure in this 2-complex in Example 3.9. It does not seem to be possible to deduce the nontriviality of α^2 from Poincaré duality alone, except when $m = 2$.

The cup product structure for an infinite-dimensional lens space L^∞ follows from the finite-dimensional case since the restriction map $H^j(L^\infty;\mathbb{Z}_m) \to H^j(L^{2n+1};\mathbb{Z}_m)$ is an isomorphism for $j \le 2n+1$. As with $\mathbb{R}P^n$, the ring structure in $H^*(L^{2n+1};\mathbb{Z})$ is determined by the ring structure in $H^*(L^{2n+1};\mathbb{Z}_m)$, and likewise for L^∞, where one has the slightly simpler structure $H^*(L^\infty;\mathbb{Z}) \approx \mathbb{Z}[\alpha]/(m\alpha)$ with $|\alpha| = 2$. The case of L^{2n+1} is obtained from this by setting $\alpha^{n+1} = 0$ and adjoining the extra $\mathbb{Z} \approx H^{2n+1}(L^{2n+1};\mathbb{Z})$.

A different derivation of the cup product structure in lens spaces is given in Example 3E.2.

Using the ad hoc notation $H^k_{free}(M)$ for $H^k(M)$ modulo its torsion subgroup, the preceding proposition implies that for a closed orientable manifold M of dimension $2n$, the middle-dimensional cup product pairing $H^n_{free}(M)\times H^n_{free}(M)\to\mathbb{Z}$ is a nonsingular bilinear form on $H^n_{free}(M)$. This form is symmetric or skew-symmetric according to whether n is even or odd. The algebra in the skew-symmetric case is rather simple: With a suitable choice of basis, the matrix of a skew-symmetric nonsingular bilinear form over $\mathbb{Z}$ can be put into the standard form consisting of 2×2 blocks $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ along the diagonal and zeros elsewhere, according to an algebra exercise at the end of the section. In particular, the rank of $H^n(M^{2n})$ must be even when n is odd. We are already familiar with these facts in the case $n = 1$ by the explicit computations of cup products for surfaces in §3.2.

The symmetric case is much more interesting algebraically. There are only finitely many isomorphism classes of symmetric nonsingular bilinear forms over $\mathbb{Z}$ of a fixed rank, but this 'finitely many' grows rather rapidly, for example it is more than 80 million for rank 32; see [Serre 1973] for an exposition of this beautiful chapter of number theory. It is known that for each even $n \ge 2$, every symmetric nonsingular form actually occurs as the cup product pairing in some closed manifold M^{2n}. One can even take M^{2n} to be simply-connected and have the bare minimum of homology: $\mathbb{Z}$'s in dimensions 0 and $2n$ and a $\mathbb{Z}^k$ in dimension n. For $n = 2$ there are at most two nonhomeomorphic simply-connected closed 4-manifolds with the same bilinear form. Namely, there are two manifolds with the same form if the square $\alpha \smile \alpha$ of some $\alpha \in H^2(M^4)$ is an odd multiple of a generator of $H^4(M^4)$, for example for $\mathbb{C}P^2$, and otherwise the M^4 is unique, for example for S^4 or $S^2\times S^2$; see [Freedman & Quinn 1990]. In §4.C we take the first step in this direction by proving a classical result of J. H. C. Whitehead that the homotopy type of a simply-connected closed 4-manifold is uniquely determined by its cup product structure.

Other Forms of Duality

Generalizing the definition of a manifold, an ***n*-manifold with boundary** is a Hausdorff space M in which each point has an open neighborhood homeomorphic either to $\mathbb{R}^n$ or to the half-space $\mathbb{R}^n_+ = \{\,(x_1,\cdots,x_n) \in \mathbb{R}^n \mid x_n \ge 0\,\}$. If a point $x \in M$ corresponds under such a homeomorphism to a point $(x_1,\cdots,x_n) \in \mathbb{R}^n_+$ with $x_n = 0$, then by excision we have $H_n(M, M-\{x\};\mathbb{Z}) \approx H_n(\mathbb{R}^n_+, \mathbb{R}^n_+ - \{0\};\mathbb{Z}) = 0$,